WHY TEST AND MANAGE

An agent is a product. Products need a test bench and an owner.

Testing answers “does it work.” **Managing** answers “who changes it, who can use it, and what it costs.” Both belong to the maker; both belong to operations.

IN THE TEST PANE

Validate conversations

Walk the agent through real questions. Watch what it picks, what it cites, where it pivots to a fallback topic.

IN PRODUCTION

See real sessions

Telemetry, session insights, success rates — the agent’s behaviour at scale, not under a lamp.

ACROSS ENVIRONMENTS

Move it deliberately

Dev, test, prod — each with rules. ALM, governance, monitoring — the rotation that keeps the product trustworthy.

TEST PERSPECTIVE

A conversation that passes once can still fail on Friday.

MANAGEMENT PERSPECTIVE

Trust is built by who owns rollback, not who made the agent.

SHARED REALITY

Both halves answer the same question: should I trust this today?

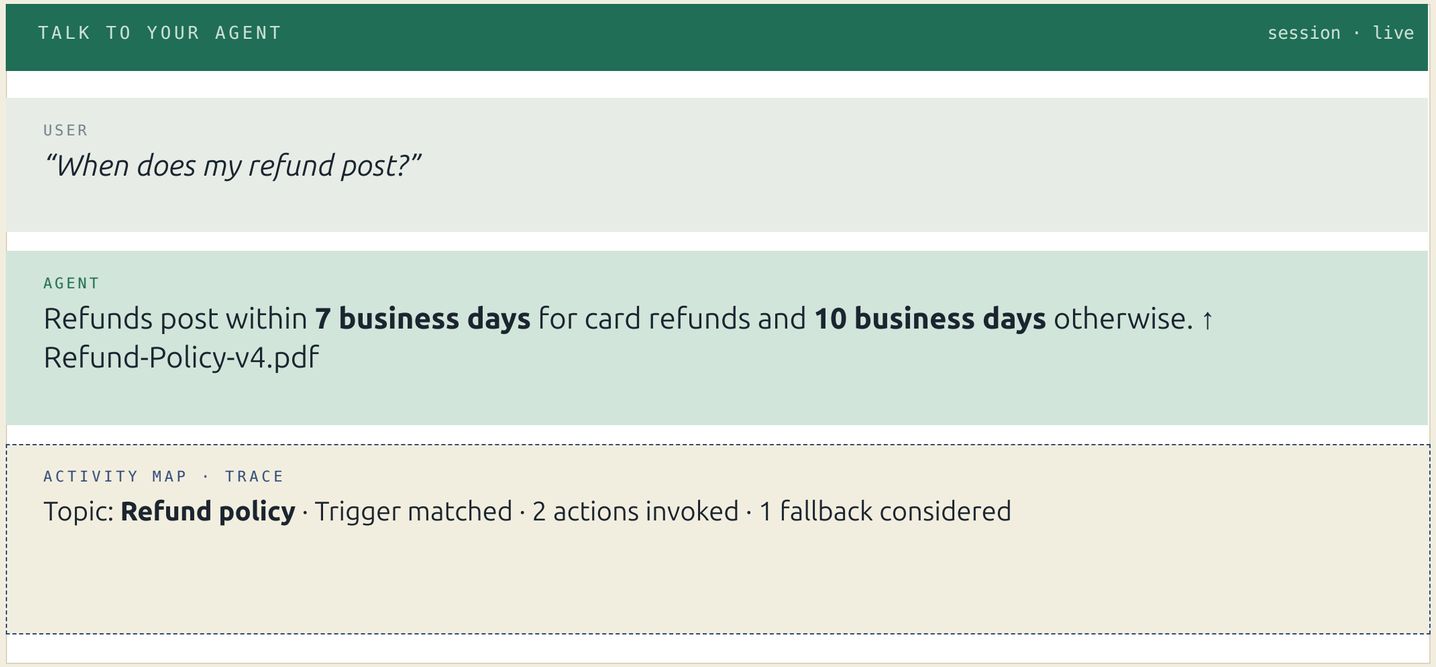
A WORKING PRODUCT, NOT A WORKING DEMO

Test once, deploy once — then keep doing both. Operational maturity beats first-launch polish.

TEST PANE · WHAT IT IS

The test pane is the agent running live, in a window.

The pane is a chat alongside the canvas. You type, the agent answers — the same engine, the same routing, the same tools. What changes is what the pane **lets you see**.



Test pane is for **walking through behaviour**. Analytics is for watching it at scale. You need both; they overlap less than you think.

WHAT THE PANE REVEALS

The surface, the route, and what it touched.

TRACES

Activity map shows which topics and tools fired.

VARIABLES

Everything the agent collected, named, filled.

CITATIONS

Source, link, snippet — the wire back to evidence.

SAVED SESSION

One-click into an evaluation set, no copy-paste.

CONVERSATION VALIDATION

A passing answer is one that holds up to all four questions at once.

A conversation is valid when you can defend it on four axes. Skip any and you publish hope, not software.

QUESTION 1 · CORRECTNESS

Did it answer what was actually asked?

Not what it could answer, not what's true in general — what the user came with. Wrong target, wrong answer.

QUESTION 2 · GROUNDING

Can a stranger verify the answer in two clicks?

The citation has to land on a real passage someone could read. "Trust me" is a failure mode.

QUESTION 3 · BOUNDARY

Did it do only what it should have done?

Did it stay inside the answer's blast radius. The agent that overreaches is the one that gets retired.

QUESTION 4 · CLARITY

Could a tired user do something with it?

An answer that's correct but un-actionable is a file the agent will slowly get trained to ignore.

Test with five real users, not with five of your own questions. The test pane won't find what you already know.

ANALYTICS · AT A GLANCE

The Monitor page is the agent's production instrumentation.

Aggregated, sampled, and scoped to a single agent. It tells you how often the agent is used, where it stalls, and what the user felt at the end of a session.

SESSIONS

How many, how long, how often

Engagement · trend lines.

RESOLUTION

Did the user leave with an answer

Resolved · escalated · abandoned.

ABANDONMENT

Where the agent loses the user

Step-by-step drop-off · per topic.

SENTIMENT

A sampled read of the room

Plus / neutral / negative.

Apply Insights — optional, deeper — gives per-message traces and richer queries. The Monitor page answers "is it's fine"; App Insights answers "which turn, why, for whom."

WHAT THIS IS NOT

It is not the truth.
It is a sampled hint.

LIMITS WORTH KNOWING

Sentiment uses a sample. Resolution depends on how the topic declares success. Abandonment is what it sounds like — read with care.

TRUST THE TREND, NOT THE DAY

Trends over weeks decide policy; one busy Tuesday does not. Pair the page with Application Insights for the trace-level story.

WHERE IT LIVES

The Monitor page aggregates an agent's analytics; Viva Insights does the same across the tenant. Both are good; pick by question.

Watch for weeks, not for hours. A dashboard that changes daily teaches you noise; one that changes quarterly teaches you the product.

CHANNELS · THREE SURFACES

A channel is where the agent shows up. Pick on purpose.

The same agent can surface in three shapes. Each one carries its own identity, audience, and identity-management story.

CHANNEL 01 · TEAMS

Inside Microsoft Teams

A bot in chat, or a Teams app. Auth follows the user's Teams identity; files respect the user's permissions.

WHEN

Work conversations already live here.

CHANNEL 02 · WEB

A standalone site

An iframe or a full canvas, embedded in your product or pointed to directly. Auth chosen by your tenant.

WHEN

The agent is the experience, not a side panel.

CHANNEL 03 · CUSTOM

Embedded in your surface

Slack, a kiosk, a support rep's console. Direct Line or your own widget; you wire auth and rendering.

WHEN

Teams is the wrong shape for the audience.

THE AGENT IS THE SAME; THE WRAPPER DIFFERS

Pick a channel by the question "where does the audience already live?" — not by which is easiest to enable.

WHERE EACH FITS

Pick the channel by audience — not by what’s easier to wire.

Three questions narrow it. The audience, the trust boundary, and what auth story you can live with.

THREE QUESTIONS

Where the audience lives

Q1 · WHERE THEY LIVE

Teams chat, your product, or a third-party surface.

Q2 · WHAT TRUST YOU OWE

Tenant identity OK, or do they need a stranger’s?

Q3 · WHO OWNS THE CHROME

Microsoft, you, or your customer.

ANSWER

The right channel falls out; defend it in writing.

IF IT’S A TEAMS ANSWER

Publish to Teams, share in a chat or as an app

Auth inherits from the user’s session; files honour the user’s permissions.

IF IT’S YOUR OWN SITE

Embed, or stand alone at a custom URL

Auth you choose; agent is the experience, not a sidekick.

IF IT’S SOMEWHERE ELSE

Wire up Direct Line or your own widget

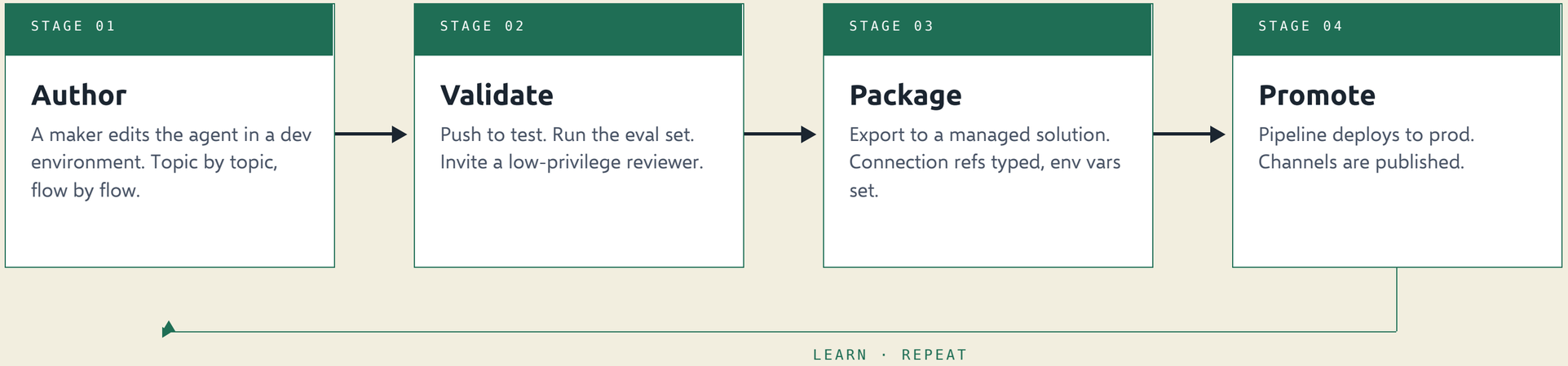
Custom rendering, custom auth — cost it before the build.

You can ship in more than one channel. Pick one first; the others earn their slot from the first one’s success.

DEPLOYMENT · THE CYCLE

A deployment is a rotation, not a moment. It’s the loop that builds the trust.

Author on a dev environment, validate on test, expose on prod — then learn from what’s there and start the next loop.



STAGE 05 · OBSERVE (ALWAYS RUNNING)

Watch the production data, then start again

Sessions, abandonment, sentiment, drift in tools and knowledge. What you see informs the next authoring round.

always on

ALM · MANAGED FIRST

A solution is the package the agent travels in.

A **solution** collects everything the agent needs: topics, flows, environment variables, connection references, knowledge sources. The agent is a row inside one.

UNMANAGED

Editable, in-progress, used for dev work.

MANAGED

Locked at import — updates add or extend, never delete.

Keep dev work in **unmanaged**; ship via **managed**.

One agent, two solutions — one truth.

WHAT'S INSIDE

Six things; one package.

01 THE AGENT

Topics, knowledge, tools, flows — one row.

02 CONNECTION REFERENCES

Typed pointers, not the secret itself.

03 ENVIRONMENT VARIABLES

Per-environment values, set on import.

04 CLOUD FLOWS

Agent flows travel with the agent.

05 MAKER-OWNED TABLES

Anything the agent writes to lives here.

06 PUBLISHER INFO

Who shipped it; who manages updates.

ONE THING UNMANAGED-VS-MANAGED CHANGES

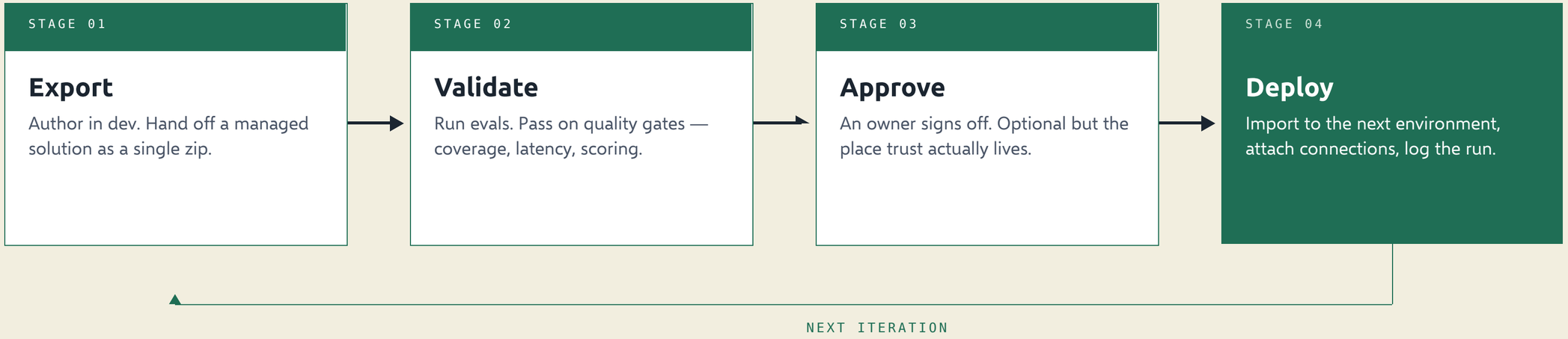
Updates after managed install add or extend. They do not delete.

Plan additions, not removals. Removing a component in prod requires a different lifecycle — uninstall, hot-fix, full redeploy. Anything else is a workaround.

PIPELINES · FROM STAGE TO STAGE

Pipelines are the rotation that replace copy-paste across envs.

A **Power Platform pipeline** deploys a managed solution from one environment to the next. It runs on demand or on a schedule, hands off approvals, and writes a record.



WHAT A PIPELINE IS NOT

A pipeline is not the agent. It is the conveyor belt the agent rides.

It deploys and configures; it does not author. Pair it with source control or ALM guidance — don’t ask it to be the workflow.

PRINCIPLE

Every environment should run the same script; never the same person.

PACKAGING · WHAT TRAVELS

Pack the solution like you'd pack a real product.

A package that ships well travels light, travels in one piece, and reads identically on every environment. Five habits earn that.

01 · ONE SOLUTION, ONE PURPOSE

Don't mix the agent with unrelated apps

Split when lifecycle or ownership differs. Two solutions are fine; one soup is not.

02 TYPED, DEFAULTED

Connections and env vars

03 EXPORT-VERSIONED

Every export stamps a version

04 DOCUMENTED

What this solution is, who owns it

05 TESTED

Before it leaves dev, ever

THE RULE

If the package can't be replayed by another person on another day, the package isn't finished.

Hard questions: who imports this without you? Where does each variable default? What happens if a connection isn't granted?

A PACKAGED AGENT, SCHEMATICALLY

A single zip with a name and a dispatch list.

Helpdesk-Bot_v3.0.2.zip — managed · 4.2 MB

ROWS KEPT INSIDE

botcomponent/ agent topics, knowledge	agent · the row
connectionreferences/ 5 connectors	typed pointers
environmentvariables/ 12 named values	set on import
cloudflows/ 2 agent flows	bundled with it
tables/ 3 maker-owned tables	data the agent writes

+ a manifest describing everything.

GOVERNANCE · WHO DECIDES

A working agent has four roles behind it, or one of them misbehaves.

A maker authors. A reviewer signs off. An owner keeps the calendar. An admin holds the keys. Handoffs between these four roles are where governance fails or holds.

ROLE · 01

Maker

Authors the agent. Owns the topic graphs, the flows, the prompts. Reads the test pane, fixes what they find. **Methods** up to local deploys.

ROLE · 02

Reviewer

Validates the agent before promotion. Runs the eval set, signs the pipeline stage, gates the prod deploy. **Stands between** dev and prod.

ROLE · 03

Owner

Carries the business outcome the agent serves. Watches analytics, owns the roadmap, raises breach alerts. **Calls the cadence** of review.

ROLE · 04

Admin

Holds the keys — environments, DLP, security roles, retention, credit caps. **Can take the agent down** faster than anyone else.

One person can hold two roles. No one should hold all four.

MONITORING · WHAT TO WATCH

A few signals earn the page; everything else is noise.

Resist the urge to dashboard everything. Four signals are enough for most agents; pick them up front and pin them.

SIGNAL 1

Resolution rate, by intent

Resolved, escalated, abandoned — broken down by topic. The trend line is the news, not the absolute.

SIGNAL 2

Tool and topic error rate

Where the agent trips — tools that 500, topics that fall through. Read together with the trace, not the dashboard.

SIGNAL 3

Cost and credit burn

Per-session tokens, per-month credits against cap. A spike here is usually a topic loop, not a real user.

SIGNAL 4

Negative feedback and escalations

A sampled thumbs-down is a story. An escalation is — usually — a sample of one that still matters.

WHAT TO SKIP

Sixteen charts no one opens is theatre, not monitoring.

If you cannot act on a signal, do not pay to display it. Action-able signals earn the column; everyone else is decoration.

MAINTENANCE · THE DRIFT YOU DON’T SEE

Maintenance is what keeps the agent honest as its world changes.

APIs drift. Knowledge ages. Models retire. A working agent stops working in months, not years. Five routines keep it’s working.

01 · WEEKLY

Read the analytics — resolution rate, escalations, abandonments by topic

Twenty minutes. The point of the weekly review is to leave with one decision.

02 · BI-WEEKLY

Re-run the eval set, including ten fresh questions nobody’s seen

Stale questions let quality drift hide. The newest ten live in your head, not in the agent.

03 · MONTHLY

Refresh knowledge sources — retire stale docs, add new ones

Citations carry the answer’s credibility; the corpus needs the same cadence the page does.

04 · QUARTERLY

Re-test tools, contracts, rate limits, model retirements

A model is auto-upgraded for 30 days — after that, your decision.

05 · WHEN THE ROLE ENDS

Retire deliberately — communicate, archive, remove

A TRICK THAT HOLDS

Write the date of the next review on the runbook.

If the next review is “sometime”, it never happens.

NAMED OWNER

Each routine needs a name, not a team.

TELEMETRY · THREE LAYERS

Telemetry is the spine between the agent and the people who run it.

Copilot Studio emits events in two scopes. Application Insights catches the trace-level story. Three layers, each one earning a different audience.

LAYER 1

Events

Discrete actions — a topic fired, a tool called, a variable set. The raw material.

who reads it · ops & analytics

Where in the loop, how often and how long, did it run?

LAYER 2

Traces

A whole session stitched together — one user, one journey, span it end-to-end through Application Insights.

who reads it · maker & reviewer

What route did the agent take, and where did it get stuck?

LAYER 3

Sessions

Aggregated, sampled, scoped to the agent or the tenant. This is what the Monitor page reads.

WHO READS IT · OWNER & ADMIN

Trend over weeks is the answer; one busy Tuesday is noise.

Wire telemetry at publish time, not at incident time.

LAB · TEST, PUBLISH, DEPLOY

Run an agent through the loop. Then watch the telemetry.

Eight steps over ~80 minutes. Test until it's boring, then publish, then read what the data says.

01 — 10 MINUTES
Walk the test pane
Ten questions, two of them unhelpful, three adversarial.

02 — 10 MINUTES
Build an eval set
20 questions, mostly real, three obviously wrong.

03 — 10 MINUTES
Publish to Teams
Flip the channel on; share in a chat with one peer.

04 — 10 MINUTES
Package as managed
Add to a solution; mark managed; export.

05 — 10 MINUTES
Run a pipeline stage
Deploy to test, tag an approval, watch it land.

06 — 10 MINUTES
Promote to prod
Same pipeline, no shortcuts; log the run.

07 — 10 MINUTES
Read the analytics page

08 — 10 MINUTES
Schedule the next review

SUCCESS SIGNALS

What "running well" actually looks like.

SIGNAL 1
Eval set passes at the gate
No regressions on the original 20; three known-weak questions are still weak.

SIGNAL 2
A peer can read the trace
Without a hand-off, the route is recoverable from telemetry.

SIGNAL 3
Review date on the runbook
Not "sometime" — the calendar has it.

SIGNAL 4
Rollback isn't a hot-fix
The pipeline can return the env to last good; no emergency surgery.

THE RULE

Trust comes from being boring on Tuesday and Friday.